

Cloud Computing

Task 4: CI/CD and Jenkins Pipeline

Objective

This assignment aims to help students practice:

- Using Git for version control
- Implementing a CI pipeline using Jenkins

Requirements

1. Unit Testing

- Write a simple program (e.g., calculator, string utility, etc.) using any language.
- Implement a few unit tests for your code using any testing framework (e.g., JUnit).

2. Git Repository

- Initialize a Git repository for your project.
- Push your source code to a remote repository (e.g., GitHub, GitLab, or Bitbucket).
- Make sure your repository is accessible.

3. Jenkins Pipeline

- Create a Jenkins pipeline script (Jenkinsfile).
- Your pipeline should include the following stages:
 - * Clone/Pull Repository
 - * Build project (optional depending on setup)
 - * Run Unit Tests
- Ensure the pipeline executes successfully and displays results.

Submission Instructions

Submit your work via Google Classroom according to the announced deadline.

Submission Format

Upload a single ZIP file named as follows:

ID1-ID2-ID3-ID4-ID5.zip

The ZIP file must contain:

- Jenkins pipeline script (Jenkinsfile)
- Link to your Git repository (in a text file or document)
- Screenshot(s) showing successful pipeline execution and test results

Notes

- Ensure your code is clean and well-structured.
- Verify that the pipeline runs without errors before submission.
- Missing any required component may affect your grade.